Chapters

Chapter	**Page**
1. Intro.	2.
2. Tetravigesimal Soyga Matrix.	3.
3. The Collatz Conjecture through the Soyga matrix.	6.
4. Recurrance of Collatz.	8.
5. The esoteric view.	8.

"Aldaraia sive Soyga vocor."

Intro.

Roughly translated, this opening line stands for: "**I am called Aldaraia or Soyga.**" It originates from the *Book of Soyga*, a 16th-century manuscript centered around cryptography, astrology, angelology, and magic. Its true origins remain largely unknown.

Most of academia analyzes the manuscript purely through what is physically written and documented. Consequently, mainstream scholars classify it strictly as a 16th-century product of European mysticism and occultism. Most literature focuses heavily on decoding the literal words and their immediate meanings, asking,, for example, what specific keywords like *NISROX* mean. While humanity has successfully reverse-engineered the mathematical mechanics behind the tables, the exact linguistic source of the keywords and the ultimate purpose of the book remain a mystery.

Let's look closer at what we *do* know. First, the text is written in Latin, which traditionally used a 23-letter alphabet (excluding J, U, and W). Second, we can spot distinct Eastern roots within the text. A prime example is the keyword *NISROX*, which strongly echoes *Nisroch*, an ancient Babylonian deity. While critics might dismiss *NISROX* as merely a meaningless "barbarian name," history shows that words and names mutate across centuries. This Babylonian link is further supported by the book's heavy emphasis on the 28 Mansiones Lunae (Lunar Mansions), an astrological concept deeply rooted in Near Eastern traditions.

Therefore, it is highly logical that the *Book of Soyga* is a Latin splinter of a far older, forgotten system. While the outer interface is Roman, the inner workings point toward ancient Hermeticism. A 36x36 grid structure creates mathematical friction when forced into a 23-letter Latin system, yet a 24-letter framework fits the underlying geometry perfectly. This structural harmony of 24 can be found in the Greek alphabet, the Elder Futhark, or the Coptic script after the Ptolemaic dynasty.

The question remains: if the system inherently requires 24 letters, why is it written in Latin? The answer lies in a hidden chain of custody across the ancient Mediterranean. During the Punic Wars, Rome waged a brutal war of expansion against Carthage. Meanwhile, Greece, Carthage, and Egypt were at peace with one another. This peace allowed esoteric knowledge to flow from the Middle East into Europe along a path through North Africa. Hannibal conquered much of Spain (Iberia) during these wars, allowing Carthaginian culture to settle deeply in the region.

Eventually, Rome defeated and conquered Carthage, Greece, and Egypt. The only place this specific knowledge could survive was in Spain, or within the cultural treasures preserved by the Moors. Historically, the Romans even gifted the literary loot of Carthage to their Moorish and Numidian allies.

Much later, Islam spread via the Umayyad Caliphate, which went on to conquer Spain. When the Umayyads discovered these ancient scriptures preserved by the Moors and Carthaginians, they brought them to intellectual hubs like the libraries of Córdoba and Toledo. Under Umayyad rule, science and philosophy flourished in Spain—the physical and

intellectual remains of which we can still see to this day. It was only when the Christian *Reconquista* occurred and Umayyad rule broke down that this safely guarded knowledge

finally spread to the rest of Europe.

So far, this appears to be the most logical conclusion of how this information came to be in the first place. However, we cannot know for certain, as no contemporary scribe wrote down the exact history of its transmission. It could have been a system so common at the time that scholars felt no need to document it, or conversely, a body of knowledge so highly complicated and hidden that it was intentionally kept out of the historical record.

Ultimately, the available information is limited strictly to what was physically written down. Speculating about these deeper origins (no matter how logical the hypotheses may be) is something academia mostly avoids due to a fear of inaccuracy or being proven wrong. Yet, we must remember that the historical record is far from flawless,, even the monks who preserved, translated, and copied these texts made frequent mathematical mistakes, as modern cryptanalysis has later revealed.

Tetravigesimal Soyga Matrix.

So what am I going to do? I will try out the formula with the Greek language, it seems that hasn't been done before.
So first let's write down how it even works.

$X=N+f(W)(\text{mod}24)$ $X=W+f(W)(\text{mod}24)$

These are the first rules, the first row is different from the second row because the consecutive row is calculated with the previous row
$X=N+f(W)(\text{mod}24)$

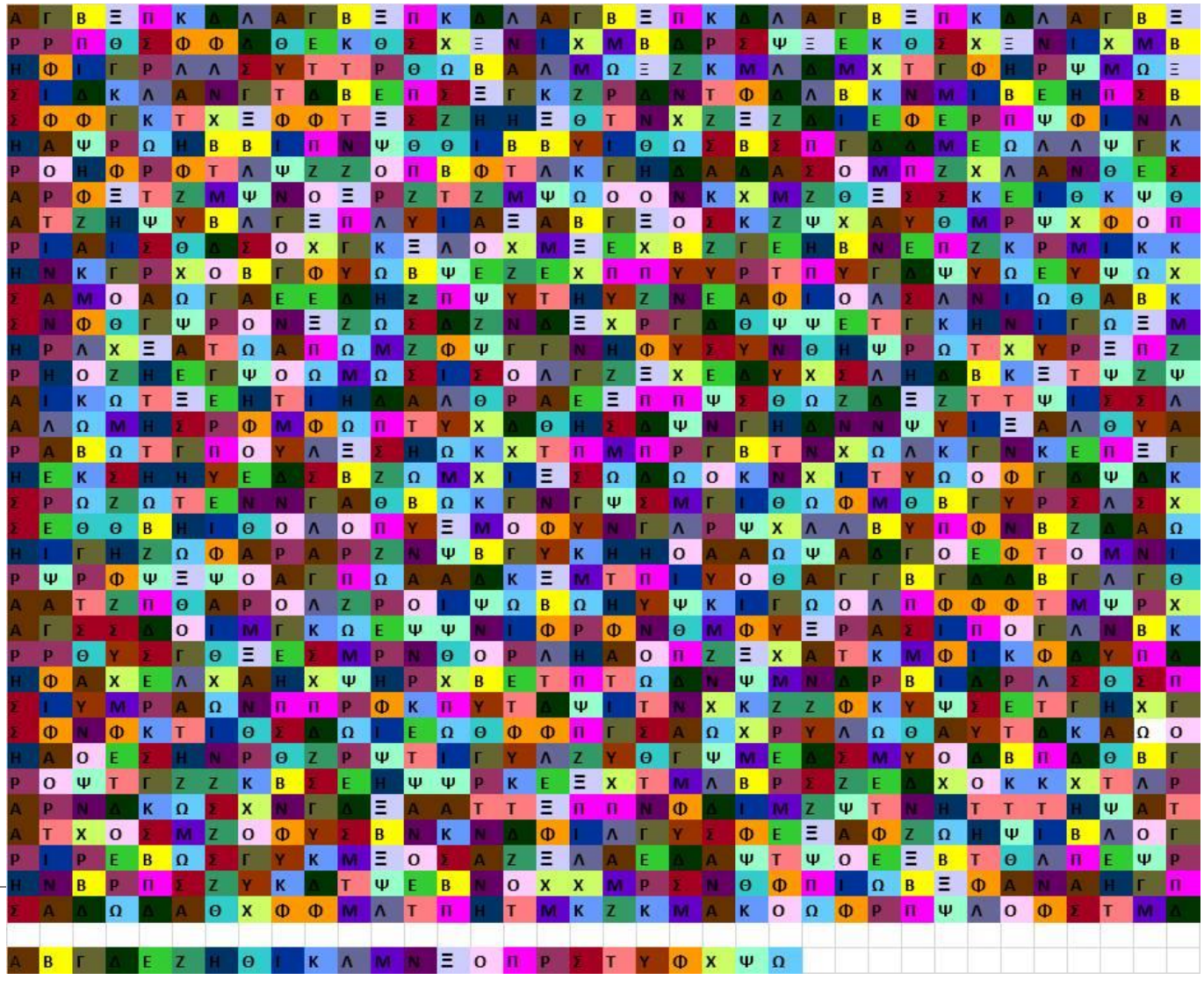
It should be noted that I count no further than the amount of letters we have, so 24 letters is the reason behind ''mod24''

Next we need a starting word and a key phrase, the key phrase needs to be the same amount as the letters, so a 24 letter sentence. I chose ''ΓΝΩΘΙ ΣΕΑΥΤΟΝ ΚΑΙ ΤΟΝ ΚΟΣΜΟΝ'' know yourself and the world. For the starting word I chose a 4 letter word because 36 divided by 4 is 9 and besides that just to be according to tradition. (The four winds and the four earthly elements present in the world of matter.) ''ΑΡΗΣ'' Ares
I have made a chart to the right so I can start making it.
After some calculations this is the first row; Α-Γ-Β-Ξ-Π-Κ-Δ-Λ,-Α.. the full result is stated below.

Letter and name	W	Key-sentence	$f(W)$
Α α — Alfa	0	**Γ γ** — Gamma	**Α α = 2**
Β β — Bèta	1	**Ν ν** — Nu	**Β β = 12**
Γ γ — Gamma	2	**Ω ω** — Omega	**Γ γ = 23**
Δ δ — Delta	3	**Θ θ** — Thèta	**Δ δ = 7**
Ε ε — Epsilon	4	**Ι ι** — Jota	**Ε ε = 8**
Ζ ζ — Zèta	5	**Σ σ (ς)** — Sigma	**Ζ ζ = 17**
Η η — Èta	6	**Ε ε** — Epsilon	**Η η = 4**
Θ θ — Thèta	7	**Α α** — Alfa	**Θ θ = 2**
Ι ι — Jota	8	**Υ υ** — Upsilon	**Ι ι = 19**
Κ κ — Kappa	9	**Τ τ** — Tau	**Κ κ = 18**
Λ λ — Lambda	10	**Ο ο** — Omikron	**Λ λ = 14**
Μ μ — Mu	11	**Ν ν** — Nu	**Μ μ = 12**
Ν ν — Nu	12	**Κ κ** — Kappa	**Ν ν = 9**
Ξ ξ — Xi	13	**Α α** — Alfa	**Ξ ξ = 2**
Ο ο — Omikron	14	**Ι ι** — Jota	**Ο ο = 8**
Π π — Pi	15	**Τ τ** — Tau	**Π π = 18**
Ρ ρ — Rho	16	**Ο ο** — Omikron	**Ρ ρ = 14**
Σ σ (ς) — Sigma	17	**Ν ν** — Nu	**Σ σ = 12**
Τ τ — Tau	18	**Κ κ** — Kappa	**Τ τ = 9**
Υ υ — Upsilon	19	**Ο ο** — Omikron	**Υ υ = 14**
Φ φ — Phi	20	**Σ σ (ς)** — Sigma	**Φ φ = 17**
Χ χ — Chi	21	**Μ μ** — Mu	**Χ χ = 11**
Ψ ψ — Psi	22	**Ο ο** — Omikron	**Ψ ψ = 14**
Ω ω — Omega	23	**Ν ν** — Nu	**Ω ω = 12**

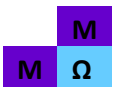
Α	Γ	Β	Ξ	Π	Κ	Δ	Λ	Α	Γ	Β	Ξ	Π	Κ	Δ	Λ	Α	Γ	Β	Ξ	Π	Κ	Δ	Λ	Α	Γ	Β	Ξ	Π	Κ	Δ	Λ	Α	Γ	Β	Ξ
Ρ	Ρ	Π	Θ	Σ	Φ	Φ	Δ	Θ	Ε	Κ	Θ	Σ	Χ	Ξ	Ν	Ι	Χ	Μ	Β	Δ	Ρ	Σ	Ψ	Ξ	Ε	Κ	Θ	Σ	Χ	Ξ	Ν	Ι	Χ	Μ	Β
Η	Φ	Ι	Γ	Ρ	Λ	Λ	Σ	Υ	Τ	Τ	Ρ	Θ	Ω	Β	Α	Λ	Μ	Ω	Ξ	Ζ	Κ	Μ	Λ	Δ	Μ	Χ	Τ	Γ	Φ	Η	Ρ	Ψ	Μ	Ω	Ξ
Σ	Ι	Δ	Κ	Λ	Α	Ν	Γ	Τ	Δ	Β	Ε	Π	Σ	Ξ	Γ	Κ	Ζ	Ρ	Δ	Ν	Τ	Φ	Δ	Λ	Β	Κ	Ν	Μ	Ι	Β	Ε	Η	Π	Σ	Β
Σ	Φ	Φ	Γ	Κ	Τ	Χ	Ξ	Φ	Φ	Τ	Ξ	Σ	Ζ	Η	Η	Ξ	Θ	Τ	Ν	Χ	Ζ	Ξ	Ζ	Δ	Ι	Ε	Φ	Ε	Ρ	Π	Ψ	Φ	Ι	Ν	Λ
Η	Α	Ψ	Ρ	Ω	Η	Β	Β	Ι	Π	Ν	Ψ	Θ	Θ	Ι	Β	Β	Υ	Ι	Θ	Ω	Σ	Β	Σ	Π	Γ	Δ	Δ	Μ	Ε	Ω	Λ	Λ	Ψ	Γ	Κ
Ρ	Ο	Η	Φ	Ρ	Φ	Τ	Λ	Ψ	Ζ	Ζ	Ο	Π	Β	Φ	Τ	Λ	Κ	Γ	Η	Δ	Α	Δ	Α	Σ	Ο	Μ	Π	Ζ	Χ	Λ	Α	Ν	Θ	Ε	Σ
Α	Ρ	Φ	Ξ	Τ	Ζ	Μ	Ψ	Ν	Ο	Ξ	Ρ	Ζ	Τ	Ζ	Μ	Ψ	Ω	Ο	Ο	Ν	Κ	Χ	Μ	Ζ	Θ	Ξ	Σ	Σ	Κ	Ε	Ι	Θ	Κ	Ψ	Θ
Α	Τ	Ζ	Η	Ψ	Υ	Β	Λ	Γ	Ξ	Π	Λ	Υ	Ι	Α	Ξ	Α	Β	Γ	Ξ	Ο	Σ	Κ	Ζ	Ψ	Χ	Α	Υ	Θ	Μ	Ρ	Ψ	Χ	Φ	Ο	Π
Ρ	Ι	Α	Ι	Σ	Θ	Δ	Σ	Ο	Χ	Γ	Κ	Ξ	Λ	Ο	Χ	Μ	Ξ	Ε	Χ	Β	Ζ	Γ	Ε	Η	Β	Ν	Ε	Π	Ζ	Κ	Ρ	Μ	Ι	Κ	Κ
Η	Ν	Κ	Γ	Ρ	Χ	Ο	Β	Γ	Φ	Υ	Ω	Β	Ψ	Ε	Ζ	Ε	Χ	Π	Π	Υ	Υ	Ρ	Τ	Π	Υ	Γ	Δ	Ψ	Υ	Ω	Ε	Υ	Ψ	Ω	Χ
Σ	Α	Μ	Ο	Α	Ω	Γ	Α	Ε	Ε	Δ	Η	z	Π	Ψ	Υ	Τ	Η	Υ	Ζ	Ν	Ε	Α	Φ	Ι	Ο	Λ	Σ	Λ	Ν	Ι	Ω	Θ	Α	Β	Κ
Σ	Ν	Φ	Θ	Γ	Ψ	Ρ	Ο	Ν	Ξ	Ζ	Ω	Σ	Δ	Ζ	Ν	Δ	Ξ	Χ	Ρ	Γ	Δ	Θ	Ψ	Ψ	Ε	Τ	Γ	Κ	Η	Ν	Ι	Γ	Ω	Ξ	Μ
Η	Ρ	Λ	Χ	Ξ	Α	Τ	Ω	Α	Π	Ω	Μ	Ζ	Φ	Ψ	Γ	Γ	Ν	Η	Φ	Υ	Σ	Υ	Ν	Θ	Η	Ψ	Ρ	Ω	Τ	Χ	Υ	Ρ	Ξ	Π	Ζ
Ρ	Η	Ο	Ζ	Η	Ε	Γ	Ψ	Ο	Ω	Μ	Ω	Σ	Ι	Σ	Ο	Λ	Γ	Ζ	Ξ	Χ	Ε	Δ	Υ	Χ	Σ	Λ	Η	Δ	Β	Κ	Ξ	Τ	Ψ	Ζ	Ψ
Α	Ι	Κ	Ω	Τ	Ξ	Ε	Η	Τ	Ι	Η	Δ	Α	Λ	Θ	Ρ	Α	Ε	Ξ	Π	Π	Ψ	Σ	Θ	Ω	Ζ	Δ	Ξ	Ζ	Τ	Τ	Ψ	Ι	Σ	Σ	Λ
Α	Λ	Ω	Μ	Η	Σ	Ρ	Φ	Μ	Φ	Ω	Π	Τ	Υ	Χ	Δ	Θ	Η	Σ	Δ	Ψ	Ν	Γ	Η	Δ	Ν	Ν	Ψ	Υ	Ι	Ξ	Α	Λ	Θ	Υ	Α
Ρ	Α	Β	Ω	Τ	Γ	Π	Ο	Υ	Λ	Ξ	Σ	Η	Ω	Κ	Χ	Τ	Π	Μ	Π	Ρ	Γ	Β	Τ	Ν	Χ	Ω	Λ	Κ	Γ	Ν	Κ	Ε	Π	Ξ	Γ
Η	Ε	Κ	Σ	Η	Η	Υ	Ε	Δ	Σ	Β	Ζ	Ω	Μ	Χ	Ι	Ξ	Σ	Ω	Δ	Ω	Ο	Κ	Ν	Χ	Ι	Τ	Υ	Ω	Ο	Φ	Γ	Δ	Ψ	Δ	Κ
Σ	Ρ	Ω	Ζ	Ω	Τ	Ε	Ν	Ν	Γ	Α	Θ	Β	Ω	Κ	Γ	Ν	Γ	Ψ	Σ	Μ	Γ	Ι	Θ	Ω	Φ	Μ	Θ	Β	Γ	Υ	Ρ	Σ	Λ	Σ	Χ
Σ	Ε	Θ	Θ	Β	Η	Ι	Θ	Ο	Λ	Ο	Π	Υ	Ξ	Μ	Ο	Φ	Υ	Ν	Γ	Λ	Ρ	Ψ	Χ	Λ	Λ	Β	Υ	Π	Φ	Ν	Β	Ζ	Δ	Α	Ω
Η	Ι	Γ	Η	Ζ	Ω	Φ	Α	Ρ	Α	Ρ	Ζ	Ν	Ψ	Β	Γ	Υ	Κ	Η	Η	Ο	Α	Α	Ω	Ψ	Α	Δ	Γ	Ο	Ε	Φ	Τ	Ο	Μ	Ν	Ι
Ρ	Ψ	Ρ	Φ	Ψ	Ξ	Ψ	Ο	Α	Γ	Π	Ω	Α	Α	Δ	Κ	Ξ	Μ	Τ	Π	Ι	Υ	Ο	Θ	Α	Γ	Γ	Β	Γ	Δ	Δ	Β	Γ	Λ	Γ	Θ
Α	Α	Τ	Ζ	Π	Θ	Α	Ρ	Ο	Λ	Ζ	Ρ	Ο	Ι	Ψ	Ω	Β	Ω	Η	Υ	Ψ	Κ	Ι	Γ	Ω	Ο	Λ	Π	Φ	Φ	Φ	Τ	Μ	Ψ	Ρ	Χ
Α	Γ	Σ	Σ	Δ	Ο	Ι	Μ	Γ	Κ	Ω	Ε	Ψ	Ψ	Ν	Ι	Φ	Ρ	Φ	Ν	Θ	Μ	Φ	Υ	Ξ	Ρ	Α	Σ	Ι	Π	Ο	Γ	Λ	Ν	Β	Κ
Ρ	Ρ	Θ	Υ	Σ	Γ	Θ	Ξ	Ε	Σ	Μ	Ρ	Ν	Θ	Ο	Ρ	Λ	Η	Α	Ο	Π	Ζ	Ξ	Χ	Α	Τ	Κ	Μ	Φ	Ι	Κ	Φ	Δ	Υ	Π	Δ
Η	Φ	Α	Χ	Ε	Λ	Χ	Α	Η	Χ	Ψ	Η	Ρ	Χ	Β	Ε	Τ	Π	Τ	Ω	Δ	Ν	Ψ	Μ	Ν	Δ	Ρ	Β	Ι	Δ	Ρ	Λ	Σ	Θ	Σ	Π
Σ	Ι	Υ	Μ	Ρ	Α	Ω	Ν	Π	Π	Ρ	Φ	Κ	Π	Υ	Τ	Δ	Ψ	Ι	Τ	Ν	Χ	Κ	Ζ	Ζ	Φ	Κ	Υ	Ψ	Σ	Ε	Τ	Γ	Η	Χ	Γ
Σ	Φ	Ν	Φ	Κ	Τ	Ι	Θ	Σ	Δ	Ω	Ι	Ε	Ω	Θ	Φ	Φ	Π	Γ	Σ	Α	Ω	Χ	Ρ	Υ	Λ	Ω	Θ	Α	Υ	Τ	Δ	Κ	Α	Ω	Ο
Η	Α	Ο	Ε	Σ	Η	Ν	Ρ	Θ	Ζ	Ρ	Ψ	Τ	Ι	Γ	Υ	Λ	Ζ	Υ	Θ	Γ	Ψ	Μ	Ε	Δ	Σ	Μ	Υ	Ο	Δ	Β	Π	Δ	Θ	Β	Γ
Ρ	Ο	Ψ	Τ	Γ	Ζ	Ζ	Κ	Β	Σ	Ε	Η	Ψ	Ψ	Ρ	Κ	Ε	Ξ	Χ	Τ	Μ	Λ	Β	Ρ	Σ	Ζ	Ε	Δ	Χ	Ο	Κ	Κ	Χ	Τ	Λ	Ρ
Α	Ρ	Ν	Δ	Κ	Ω	Σ	Χ	Ν	Γ	Δ	Ξ	Α	Α	Τ	Τ	Ξ	Π	Π	Ν	Φ	Δ	Ι	Μ	Ζ	Ψ	Τ	Ν	Η	Τ	Τ	Τ	Η	Ψ	Α	Τ
Α	Τ	Χ	Ο	Σ	Μ	Ζ	Ο	Φ	Υ	Σ	Β	Ν	Κ	Ν	Δ	Φ	Ι	Λ	Γ	Υ	Σ	Φ	Ε	Ξ	Α	Φ	Ζ	Ω	Η	Ψ	Ι	Β	Λ	Ο	Γ
Ρ	Ι	Ρ	Ε	Β	Ω	Σ	Γ	Υ	Κ	Μ	Ξ	Ο	Σ	Α	Ζ	Ξ	Λ	Α	Ε	Δ	Α	Ψ	Τ	Ψ	Ο	Ε	Ξ	Β	Τ	Θ	Λ	Π	Ε	Ψ	Ρ
Η	Ν	Β	Ρ	Π	Σ	Ζ	Υ	Κ	Δ	Τ	Ψ	Ε	Β	Ν	Ο	Χ	Χ	Μ	Ρ	Σ	Ν	Θ	Φ	Π	Ι	Ω	Β	Ξ	Φ	Α	Ν	Α	Η	Γ	Π
Σ	Α	Δ	Ω	Δ	Α	Θ	Χ	Φ	Φ	Μ	Λ	Τ	Π	Η	Τ	Μ	Κ	Ζ	Κ	Μ	Α	Κ	Ο	Ω	Φ	Ρ	Π	Ψ	Λ	Ο	Φ	Σ	Τ	Μ	Δ

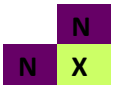
<Note human made and could pertain errors>



When making this there were a couple of thins standing out. The first was that when I made a row I recounted every letter to minimize mistakes, when I did made mistakes it was rare to lose the whole row, the code seemed to repair itself over time, sometimes even after 5 or less letters.

The other thing was the recurrence and grouping of some letters forming something I can best describe as islands or nodes as these two per example are.





There are also parts that cross like a bridge like this example scattered all around the field.



The grid behaves like a repeating wave and it makes me think of something like a seed generator in videogames.

My hypothesis is that it probably was used as an analogue mathematical simulation of the stars. this seems historically the most plausible looking at the cultures.

So what could we do with this in modern times?

Perhaps steganography, data compression or a seed generator. I should note that I used Greek language but it would also work with the futhark or Coptic. Noting also that with the Coptic 24 is necessary.

Honestly I could not think of many more things than perhaps for indie game developers or something of the like.

But I give some pointers to think about with this document, these things come to mind.

- "as a blueprint for physical transistor circuits"
- "Semantic compression for AI models"
- "the gradient explosion/vanishing problem in deep neural networks."

Perhaps there is something even more useful in this but honestly to speculate about that is more abstract than the historical speculations made already. I do hope that experts in certain fields recognize this as something useful, the one man's trash is the other man's treasure.

Without archeologic evidence this is nothing more than a plausible theory and there is a possible chance that we will never know the true origins with absolute certainty because not everything that happened in history is also written down.

The Collatz Conjecture through the Soyga matrix.

This document was meant to be finished at this point but I thought about the Collatz conjecture and I was wondering what would happen if I use the Soyga matrix on it. It seemd that though base-12 there is a known phenomenon of rhythm so I only had to look at the same thing with base-24. I figured that I could use that as a key phrase and I could use the trivial cycle as the starting word.

- **The Key Phrase** is numbered according to the Collatz base-24 rhythm:
  4, 1, 10, 2, 16, 3, 22, 4, 4, 5, 10, 6, 16, 7, 22, 8, 4, 9, 10, 10, 16, 11, 22, 12
- **The Starting Word** is the ultimate ground state of the conjecture:
  4, 2, 1. (1, 2, 4.)

By combining these two, the infinite math of Collatz is forced into the fixed, 24-character grid of the Soyga matrix. Instead of random letters, this specific combination uses a pure mathematical rhythm to drive the equations. To show the key phrase and starting word I show them below.

- The Key Phrase: Δ,A,K,B,Π,Γ,X,Δ, Δ,E,K,Z,Π,H,X,Θ,Δ,I,K,K,Π,Λ,X,M.
- The Starting Word: Δ,B,A,A,B,Δ.

Letter and name	W	$f(W)$
Α α — Alfa	1	4
Β β — Bèta	2	1
Γ γ — Gamma	3	10
Δ δ — Delta	4	2
Ε ε — Epsilon	5	16
Ζ ζ — Zèta	6	3
Η η — Èta	7	22
Θ θ — Thèta	8	4
Ι ι — Jota	9	4
Κ κ — Kappa	10	5
Λ λ — Lambda	11	10
Μ μ — Mu	12	6
Ν ν — Nu	13	16
Ξ ξ — Xi	14	7
Ο ο — Omikron	15	22
Π π — Pi	16	8
Ρ ρ — Rho	17	4
Σ σ (ς) — Sigma	18	9
Τ τ — Tau	19	10
Υ υ — Upsilon	20	10
Φ φ — Phi	21	16
Χ χ — Chi	22	11
Ψ ψ — Psi	23	22
Ω ω — Omega	24	12

The reason why I chose this approach is because a word is a choice, but a sequence is a fact. This way, perhaps a more scientific conclusion can arrive from this. I will look later at the esoteric connections, instead of taking them as the source.

$X=N+f(W)(\text{mod}24)$ $X=W+f(W)(\text{mod}24)$

The thing that is most on my mind with this subject is that we know how it works but not what it is used for or how it is useful. How much I would like to say that I have found a hidden law or some other form of conclusion, I simply cannot.

It could be my lack of knowledge so I leave it at this, I hope that mathematicians and experts on the Collatz conjecture see something useful in this data.

On the page below I have noted the result and I improved the choice of colour to make it a bit more readable.

Δ	Ζ	Ι	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ
Β	Η	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ
Α	Λ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ	Φ	Τ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ	Φ	Τ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ
Α	Ο	Ο	Ν	Ε	Γ	Τ	Ρ	Ι	Η	Θ	Γ	Η	Ρ	Φ	Η	Λ	Φ	Α	Λ	Ο	Α	Ε	Ο	Τ	Ε	Ι	Τ	Ψ	Ι	Ν	Ψ	Γ	Ν	Ρ	Γ
Β	Π	Ψ	Λ	Ο	Α	Ψ	Ο	Η	Ε	Ω	Ο	Ε	Ι	Α	Λ	Φ	Ν	Ρ	Ο	Ν	Ρ	Ι	Τ	Ε	Φ	Α	Ψ	Φ	Α	Ρ	Γ	Ν	Ε	Ι	Η
Δ	Σ	Θ	Ο	Ν	Ρ	Γ	Α	Λ	Ο	Χ	Β	Ζ	Μ	Η	Ι	Α	Ρ	Τ	Α	Ρ	Φ	Α	Ψ	Γ	Η	Ψ	Φ	Ν	Ρ	Τ	Ν	Ε	Φ	Α	Λ
Δ	Υ	Σ	Ω	Α	Φ	Τ	Λ	Φ	Η	Υ	Μ	Μ	Σ	Π	Ρ	Ε	Ι	Ψ	Ψ	Ο	Ε	Ρ	Γ	Ν	Ψ	Φ	Ν	Ε	Ι	Ψ	Λ	Ο	Τ	Λ	Φ
Β	Φ	Κ	Ε	Ρ	Α	Ψ	Ι	Α	Ψ	Σ	Φ	Δ	Υ	Β	Σ	Ξ	Π	Η	Φ	Η	Γ	Γ	Ν	Ε	Ο	Τ	Ψ	Γ	Τ	Ι	Ο	Ν	Λ	Φ	Ν
Α	Α	Ξ	Μ	Ψ	Ψ	Φ	Α	Ε	Ο	Π	Ε	Υ	Ζ	Ε	Κ	Τ	Β	Θ	Α	Λ	Ν	Τ	Ψ	Γ	Α	Ψ	Φ	Τ	Ε	Α	Τ	Ψ	Ι	Α	Ρ
Α	Ε	Ζ	Ο	Φ	Ο	Τ	Λ	Ο	Ν	Θ	Ι	Ω	Σ	Ξ	Ρ	Ψ	Ω	Υ	Λ	Φ	Ε	Λ	Ι	Η	Ψ	Φ	Ν	Λ	Ο	Ψ	Ρ	Γ	Τ	Λ	Γ
Β	Ζ	Ι	Τ	Η	Ν	Λ	Φ	Η	Λ	Σ	Σ	Ι	Χ	Α	Φ	Ο	Χ	Η	Ι	Α	Ι	Ο	Η	Ε	Ο	Τ	Ψ	Ι	Τ	Ι	Φ	Τ	Ε	Γ	Ν
Δ	Θ	Ν	Λ	Ρ	Ρ	Ο	Ε	Ψ	Ι	Χ	Ε	Α	Β	Β	Χ	Β	Ψ	Ε	Α	Ε	Α	Τ	Β	Ζ	Σ	Δ	Α	Ν	Λ	Τ	Η	Ρ	Ι	Η	Λ
Δ	Κ	Σ	Υ	Γ	Γ	Α	Ι	Γ	Τ	Θ	Ι	Ε	Σ	Λ	Θ	Ζ	Β	Ζ	Δ	Η	Ψ	Φ	Σ	Ο	Π	Μ	Η	Λ	Φ	Λ	Ρ	Φ	Α	Λ	Φ
Β	Λ	Δ	Χ	Ξ	Κ	Ζ	Μ	Ι	Ψ	Ζ	Μ	Λ	Δ	Ν	Ω	Σ	Λ	Π	Μ	Ν	Ο	Τ	Δ	Ρ	Υ	Χ	Σ	Υ	Η	Ι	Φ	Ν	Ρ	Ο	Τ
Α	Ο	Β	Ψ	Μ	Π	Ξ	Τ	Τ	Ι	Κ	Ρ	Ο	Β	Ξ	Η	Π	Τ	Β	Ν	Ε	Η	Ρ	Θ	Φ	Μ	Δ	Υ	Ζ	Κ	Ξ	Δ	Ο	Ο	Ν	Λ
Α	Τ	Μ	Ε	Δ	Σ	Ψ	Ρ	Ψ	Η	Θ	Φ	Η	Ω	Β	Θ	Υ	Ε	Σ	Χ	Π	Ο	Ο	Ζ	Ω	Ω	Π	Δ	Θ	Ξ	Φ	Υ	Α	Τ	Ψ	Ι
Β	Υ	Χ	Π	Μ	Ω	Λ	Γ	Ι	Λ	Σ	Ζ	Κ	Ε	Σ	Ρ	Ω	Ρ	Χ	Ι	Υ	Α	Τ	Π	Θ	Δ	Σ	Ν	Ω	Β	Χ	Η	Ψ	Ρ	Γ	Τ
Δ	Ω	Κ	Φ	Δ	Β	Μ	Ι	Ν	Γ	Δ	Θ	Ξ	Μ	Ω	Ε	Π	Α	Β	Κ	Α	Ε	Λ	Β	Ι	Θ	Χ	Ω	Μ	Θ	Β	Θ	Γ	Γ	Ν	Λ
Δ	Β	Λ	Η	Β	Γ	Χ	Υ	Ψ	Α	Θ	Μ	Υ	Χ	Λ	Ο	Ξ	Θ	Ζ	Ν	Ρ	Ι	Ο	Ω	Φ	Ω	Κ	Ε	Δ	Κ	Η	Ζ	Ζ	Ζ	Π	Τ
Β	Γ	Φ	Ψ	Ω	Ο	Υ	Ζ	Β	Β	Ι	Π	Δ	Ω	Ψ	Ν	Ζ	Λ	Π	Φ	Ι	Ν	Η	Χ	Θ	Δ	Μ	Λ	Ξ	Ρ	Λ	Π	Ξ	Ν	Θ	Ψ
Α	Η	Τ	Ι	Δ	Ρ	Ω	Σ	Λ	Μ	Ο	Ξ	Λ	Κ	Δ	Ο	Δ	Ν	Θ	Α	Ν	Ε	Ψ	Υ	Σ	Ν	Δ	Ν	Ζ	Υ	Φ	Θ	Σ	Χ	Τ	Ι
Α	Λ	Ε	Α	Θ	Φ	Π	Β	Ρ	Π	Ψ	Μ	Π	Σ	Ν	Η	Β	Ξ	Ο	Ψ	Λ	Ο	Φ	Μ	Ω	Α	Θ	Ρ	Κ	Α	Α	Μ	Ω	Κ	Ω	Φ
Β	Μ	Λ	Μ	Ξ	Δ	Σ	Λ	Γ	Β	Ω	Ω	Δ	Υ	Ψ	Ε	Σ	Ψ	Ν	Ο	Ι	Τ	Η	Κ	Ε	Ρ	Μ	Ψ	Θ	Ε	Ρ	Π	Θ	Ξ	Η	Τ
Δ	Ξ	Σ	Φ	Ζ	Η	Π	Τ	Ν	Σ	Ι	Δ	Ζ	Ψ	Φ	Φ	Κ	Δ	Ο	Ν	Α	Ψ	Ε	Β	Ζ	Υ	Χ	Κ	Ν	Φ	Ι	Υ	Σ	Ψ	Ε	Λ
Δ	Π	Β	Χ	Ρ	Λ	Β	Υ	Ψ	Π	Ρ	Θ	Κ	Δ	Ψ	Τ	Υ	Ξ	Χ	Ω	Ν	Ο	Γ	Μ	Μ	Β	Ψ	Θ	Ρ	Α	Ν	Μ	Ω	Λ	Ο	Ι
Β	Ρ	Ζ	Α	Φ	Γ	Μ	Β	Ω	Δ	Τ	Σ	Τ	Ξ	Ζ	Χ	Η	Μ	Δ	Β	Ξ	Χ	Ξ	Τ	Χ	Ν	Ο	Ζ	Υ	Λ	Ψ	Κ	Ε	Γ	Α	Ν
Α	Φ	Χ	Μ	Γ	Ν	Δ	Δ	Β	Ε	Λ	Δ	Φ	Ζ	Ι	Β	Θ	Π	Μ	Θ	Σ	Η	Μ	Α	Β	Ξ	Χ	Ρ	Ω	Ψ	Φ	Β	Ζ	Ζ	Δ	Ο
Α	Α	Β	Ν	Τ	Ψ	Β	Ε	Σ	Ξ	Σ	Ν	Ν	Χ	Υ	Μ	Ξ	Ψ	Κ	Ν	Κ	Μ	Σ	Κ	Η	Μ	Δ	Τ	Λ	Ι	Α	Ζ	Ι	Κ	Ι	Τ
Β	Β	Γ	Ψ	Ρ	Γ	Μ	Λ	Δ	Π	Β	Ξ	Υ	Θ	Ω	Ω	Β	Ω	Χ	Ω	Χ	Ψ	Π	Σ	Π	Υ	Ξ	Β	Μ	Ο	Ψ	Δ	Λ	Υ	Τ	Ε
Δ	Δ	Ε	Ο	Ο	Α	Π	Τ	Ξ	Ψ	Ω	Β	Φ	Ω	Μ	Ζ	Ε	Π	Ζ	Γ	Θ	Γ	Β	Τ	Β	Φ	Ζ	Ε	Δ	Ρ	Γ	Ξ	Σ	Ε	Λ	Ο
Δ	Ζ	Θ	Τ	Α	Ε	Θ	Ψ	Μ	Ε	Π	Κ	Β	Α	Π	Ξ	Μ	Χ	Ρ	Η	Ζ	Ζ	Ε	Λ	Μ	Γ	Π	Ν	Υ	Γ	Ν	Ζ	Φ	Φ	Γ	Α
Β	Η	Ζ	Χ	Μ	Λ	Σ	Θ	Π	Ν	Θ	Ξ	Ι	Ε	Θ	Σ	Φ	Ξ	Ω	Τ	Π	Ξ	Μ	Ρ	Π	Λ	Β	Ξ	Γ	Ν	Ε	Χ	Θ	Α	Η	Ψ
Α	Λ	Π	Ζ	Ο	Ι	Χ	Τ	Β	Ξ	Ο	Μ	Ο	Γ	Σ	Γ	Η	Μ	Ζ	Χ	Γ	Ω	Ω	Γ	Β	Μ	Θ	Σ	Μ	Τ	Ο	Υ	Σ	Κ	Μ	Ε
Α	Ο	Ξ	Ν	Η	Η	Υ	Ε	Σ	Ψ	Ν	Δ	Ρ	Η	Π	Λ	Ρ	Π	Ξ	Ε	Τ	Κ	Ε	Τ	Μ	Σ	Ρ	Χ	Ψ	Ρ	Τ	Ζ	Φ	Β	Ν	Φ
Β	Π	Χ	Ω	Τ	Ρ	Ω	Ρ	Χ	Α	Ρ	Θ	Φ	Ψ	Ξ	Σ	Β	Ρ	Σ	Ξ	Β	Λ	Ο	Ρ	Π	Β	Σ	Η	Φ	Ι	Ψ	Δ	Ψ	Ω	Α	Α
Δ	Σ	Η	Χ	Ζ	Υ	Κ	Χ	Ι	Ε	Ι	Μ	Γ	Ι	Σ	Γ	Μ	Ψ	Π	Χ	Ν	Γ	Α	Φ	Θ	Ζ	Φ	Ψ	Τ	Τ	Ι	Θ	Γ	Κ	Ζ	Δ

< Note human made and could pertain errors>

Δ	Ζ	Ι	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ	Ν	Ε	Φ
Β	Η	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ	Ψ	Γ	Η	Λ	Ο	Τ
Α	Λ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ	Φ	Τ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ	Φ	Τ	Ρ	Ο	Ν	Λ	Ι	Η	Ε	Γ	Α	Ψ
Α	Ο	Ο	Ν	Ε	Γ	Τ	Ρ	Ι	Η	Θ	Γ	Η	Ρ	Φ	Η	Λ	Φ	Α	Λ	Ο	Α	Ε	Ο	Τ	Ε	Ι	Τ	Ψ	Ι	Ν	Ψ	Γ	Ν	Ρ	Γ
Β	Π	Ψ	Λ	Ο	Α	Ψ	Ο	Η	Ε	Ω	Ο	Ε	Ι	Α	Λ	Φ	Ν	Ρ	Ο	Ν	Ρ	Ι	Τ	Ε	Φ	Α	Ψ	Φ	Α	Ρ	Γ	Ν	Ε	Ι	Η
Δ	Σ	Θ	Ο	Ν	Ρ	Γ	Α	Λ	Ο	Χ	Β	Ζ	Μ	Η	Ι	Α	Ρ	Τ	Α	Ρ	Φ	Α	Ψ	Γ	Η	Ψ	Φ	Ν	Ρ	Τ	Ν	Ε	Φ	Α	Λ
Δ	Υ	Σ	Ω	Α	Φ	Τ	Λ	Φ	Η	Υ	Μ	Μ	Σ	Π	Ρ	Ε	Ι	Ψ	Ψ	Ο	Ε	Ρ	Γ	Ν	Ψ	Φ	Ν	Ε	Ι	Ψ	Λ	Ο	Τ	Λ	Φ
Β	Φ	Κ	Ε	Ρ	Α	Ψ	Ι	Α	Ψ	Σ	Φ	Δ	Υ	Β	Σ	Ξ	Π	Η	Φ	Η	Γ	Γ	Ν	Ε	Ο	Τ	Ψ	Γ	Τ	Ι	Ο	Ν	Λ	Φ	Ν
Α	Α	Ξ	Μ	Ψ	Ψ	Φ	Α	Ε	Ο	Π	Ε	Υ	Ζ	Ε	Κ	Τ	Β	Θ	Α	Λ	Ν	Τ	Ψ	Γ	Α	Ψ	Φ	Τ	Ε	Α	Τ	Ψ	Ι	Α	Ρ
Α	Ε	Ζ	Ο	Φ	Ο	Τ	Λ	Ο	Ν	Θ	Ι	Ω	Σ	Ξ	Ρ	Ψ	Ω	Υ	Λ	Φ	Ε	Λ	Ι	Η	Ψ	Φ	Ν	Λ	Ο	Ψ	Ρ	Γ	Τ	Λ	Γ
Β	Ζ	Ι	Τ	Η	Ν	Λ	Φ	Η	Λ	Σ	Σ	Ι	Χ	Α	Φ	Ο	Χ	Η	Ι	Α	Ι	Ο	Η	Ε	Ο	Τ	Ψ	Ι	Τ	Ι	Φ	Τ	Ε	Γ	Ν
Δ	Θ	Ν	Λ	Ρ	Ρ	Ο	Ε	Ψ	Ι	Χ	Ε	Α	Β	Β	Χ	Β	Ψ	Ε	Α	Ε	Α	Τ	Β	Ζ	Σ	Δ	Α	Ν	Λ	Τ	Η	Ρ	Ι	Η	Λ
Δ	Κ	Σ	Υ	Γ	Γ	Α	Ι	Γ	Τ	Θ	Ι	Ε	Σ	Λ	Θ	Ζ	Β	Ζ	Δ	Η	Ψ	Φ	Σ	Ο	Π	Μ	Η	Λ	Φ	Λ	Ρ	Φ	Α	Λ	Φ
Β	Λ	Δ	Χ	Ξ	Κ	Ζ	Μ	Ι	Ψ	Ζ	Μ	Λ	Δ	Ν	Ω	Σ	Λ	Π	Μ	Ν	Ο	Τ	Δ	Ρ	Υ	Χ	Σ	Υ	Η	Ι	Φ	Ν	Ρ	Ο	Τ
Α	Ο	Β	Ψ	Μ	Π	Ξ	Τ	Τ	Ι	Κ	Ρ	Ο	Β	Ξ	Η	Π	Τ	Β	Ν	Ε	Η	Ρ	Θ	Φ	Μ	Δ	Υ	Ζ	Κ	Ξ	Δ	Ο	Ο	Ν	Λ
Α	Τ	Μ	Ε	Δ	Σ	Ψ	Ρ	Ψ	Η	Θ	Φ	Η	Ω	Β	Θ	Υ	Ε	Σ	Χ	Π	Ο	Ο	Ζ	Ω	Ω	Π	Δ	Θ	Ξ	Φ	Υ	Α	Τ	Ψ	Ι
Β	Υ	Χ	Π	Μ	Ω	Λ	Γ	Ι	Λ	Σ	Ζ	Κ	Ε	Σ	Ρ	Ω	Ρ	Χ	Ι	Υ	Α	Τ	Π	Θ	Δ	Σ	Ν	Ω	Β	Χ	Η	Ψ	Ρ	Γ	Τ
Δ	Ω	Κ	Φ	Δ	Β	Μ	Ι	Ν	Γ	Δ	Θ	Ξ	Μ	Ω	Ε	Π	Α	Β	Κ	Α	Ε	Λ	Β	Ι	Θ	Χ	Ω	Μ	Θ	Β	Θ	Γ	Γ	Ν	Λ
Δ	Β	Λ	Η	Β	Γ	Χ	Υ	Ψ	Α	Θ	Μ	Υ	Χ	Λ	Ο	Ξ	Θ	Ζ	Ν	Ρ	Ι	Ο	Ω	Φ	Ω	Κ	Ε	Δ	Κ	Η	Ζ	Ζ	Ζ	Π	Τ
Β	Γ	Φ	Ψ	Ω	Ο	Υ	Ζ	Β	Β	Ι	Π	Δ	Ω	Ψ	Ν	Ζ	Λ	Π	Φ	Ι	Ν	Η	Χ	Θ	Δ	Μ	Λ	Ξ	Ρ	Λ	Π	Ξ	Ν	Θ	Ψ
Α	Η	Τ	Ι	Δ	Ρ	Ω	Σ	Λ	Μ	Ο	Ξ	Λ	Κ	Δ	Ο	Δ	Ν	Θ	Α	Ν	Ε	Ψ	Υ	Σ	Ν	Δ	Ν	Ζ	Υ	Φ	Θ	Σ	Χ	Τ	Ι
Α	Λ	Ε	Α	Θ	Φ	Π	Β	Ρ	Π	Ψ	Μ	Π	Σ	Ν	Η	Β	Ξ	Ο	Ψ	Λ	Ο	Φ	Μ	Ω	Α	Θ	Ρ	Κ	Α	Α	Μ	Ω	Κ	Ω	Φ
Β	Μ	Λ	Μ	Ξ	Δ	Σ	Λ	Γ	Β	Ω	Ω	Δ	Υ	Ψ	Ε	Σ	Ψ	Ν	Ο	Ι	Τ	Η	Κ	Ε	Ρ	Μ	Ψ	Θ	Ε	Ρ	Π	Θ	Ξ	Η	Τ
Δ	Ξ	Σ	Φ	Ζ	Η	Π	Τ	Ν	Σ	Ι	Δ	Ζ	Ψ	Φ	Φ	Κ	Δ	Ο	Ν	Α	Ψ	Ε	Β	Ζ	Υ	Χ	Κ	Ν	Φ	Ι	Υ	Σ	Ψ	Ε	Λ
Δ	Π	Β	Χ	Ρ	Λ	Β	Υ	Ψ	Π	Ρ	Θ	Κ	Δ	Ψ	Τ	Υ	Ξ	Χ	Ω	Ν	Ο	Γ	Μ	Μ	Β	Ψ	Θ	Ρ	Α	Ν	Μ	Ω	Λ	Ο	Ι
Β	Ρ	Ζ	Α	Φ	Γ	Μ	Β	Ω	Δ	Τ	Σ	Τ	Ξ	Ζ	Χ	Η	Μ	Δ	Β	Ξ	Χ	Ξ	Τ	Χ	Ν	Ο	Ζ	Υ	Λ	Ψ	Κ	Ε	Γ	Α	Ν
Α	Φ	Χ	Μ	Γ	Ν	Δ	Δ	Β	Ε	Λ	Δ	Φ	Ζ	Ι	Β	Θ	Π	Μ	Θ	Σ	Η	Μ	Α	Β	Ξ	Χ	Ρ	Ω	Ψ	Φ	Β	Ζ	Ζ	Δ	Ο
Α	Α	Β	Ν	Τ	Ψ	Β	Ε	Σ	Ξ	Σ	Ν	Ν	Χ	Υ	Μ	Ξ	Ψ	Κ	Ν	Κ	Μ	Σ	Κ	Η	Μ	Δ	Τ	Λ	Ι	Α	Ζ	Ι	Κ	Ι	Τ
Β	Β	Γ	Ψ	Ρ	Γ	Μ	Λ	Δ	Π	Β	Ξ	Υ	Θ	Ω	Ω	Β	Ω	Χ	Ω	Χ	Ψ	Π	Σ	Π	Υ	Ξ	Β	Μ	Ο	Ψ	Δ	Λ	Υ	Τ	Ε
Δ	Δ	Ε	Ο	Ο	Α	Π	Τ	Ξ	Ψ	Ω	Β	Φ	Ω	Μ	Ζ	Ε	Π	Ζ	Γ	Θ	Γ	Β	Τ	Β	Φ	Ζ	Ε	Δ	Ρ	Γ	Ξ	Σ	Ε	Λ	Ο
Δ	Ζ	Θ	Τ	Α	Ε	Θ	Ψ	Μ	Ε	Π	Κ	Β	Α	Π	Ξ	Μ	Χ	Ρ	Η	Ζ	Ζ	Ε	Λ	Μ	Γ	Π	Ν	Υ	Γ	Ν	Ζ	Φ	Φ	Γ	Α
Β	Η	Ζ	Χ	Μ	Λ	Σ	Θ	Π	Ν	Θ	Ξ	Ι	Ε	Θ	Σ	Φ	Ξ	Ω	Τ	Π	Ξ	Μ	Ρ	Π	Λ	Β	Ξ	Γ	Ν	Ε	Χ	Θ	Α	Η	Ψ
Α	Λ	Π	Ζ	Ο	Ι	Χ	Τ	Β	Ξ	Ο	Μ	Ο	Γ	Σ	Γ	Η	Μ	Ζ	Χ	Γ	Ω	Ω	Γ	Β	Μ	Θ	Σ	Μ	Τ	Ο	Υ	Σ	Κ	Μ	Ε
Α	Ο	Ξ	Ν	Η	Η	Υ	Ε	Σ	Ψ	Ν	Δ	Ρ	Η	Π	Λ	Ρ	Π	Ξ	Ε	Τ	Κ	Ε	Τ	Μ	Σ	Ρ	Χ	Ψ	Ρ	Τ	Ζ	Φ	Β	Ν	Φ
Β	Π	Χ	Ω	Τ	Ρ	Ω	Ρ	Χ	Α	Ρ	Θ	Φ	Ψ	Ξ	Σ	Β	Ρ	Σ	Ξ	Β	Λ	Ο	Ρ	Π	Β	Σ	Η	Φ	Ι	Ψ	Δ	Ψ	Ω	Α	Α
Δ	Σ	Η	Χ	Ζ	Υ	Κ	Χ	Ι	Ε	Ι	Μ	Γ	Ι	Σ	Γ	Μ	Ψ	Π	Χ	Ν	Γ	Α	Φ	Θ	Ζ	Φ	Ψ	Τ	Τ	Ι	Θ	Γ	Κ	Ζ	Δ

Α	Β	Γ	Δ	Ε	Ζ	Η	Θ	Ι	Κ	Λ	Μ	Ν	Ξ	Ο	Π	Ρ	Σ	Τ	Υ	Φ	Χ	Ψ	Ω

Recurrance of Collatz.

As this later added part is wanted to look at the recurring patterns,

I looked at the first six rows and the results made it difficult to continue.

- The first row starts with **Δ, Z** and **I** starts after that start with a perfect loop of **N, E** and **Φ.**
- The second row starts with **B** and **H.** the loop starts here with **H, Λ,O,T, Ψ** and **Γ.**

  Here we notice the shift and see the loop getting bigger with 6 letters instead of 3.

- The third row starts with **A** and **Λ**. The same as the last. But here the loop is **P,O,N,Λ,I,H,E,Γ,A,Ψ,Φ** and **T.** this is 12. Instead of 6.
- The fourth row starts with **A,O,O,N,E,Γ,T,P,I,H** and **Θ.** so instead of a start of 2 we get a start of 11, the loop that recurs is **Γ,H,P,Φ,H,Λ,Φ,A,Λ,O,A,E,O,T,E,I,T,Ψ,I,N,Ψ,Γ,N** and **P.** So instead of 12 we have 24.
- The fifth row starts with **B,Π,Ψ,Λ,O,A,Ψ,O,H,E,Ω** and **O.** so the start is now 12 and the loop is 24 total with **E,I,A, Λ, Φ,N,P,O,N,P,I,T,E, Φ,A, Ψ, Φ,A,P, Γ,N,E,I** and **H.** coincidently is this also the total of the Soyga grid captured with the loop and the start.
- The sixth row exploded in length with a as start of **Δ,Σ,Θ,O,N,P,Γ,A,Λ,O,X,B,Z,M,H,I,A,P,T,A,P,Φ,A,Ψ,Γ,H,Ψ,Φ,N,P,T,** and **N.** So now the row is almost fully filled with the start by being 32 letters long. The loop extended far beyond the Soyga.
  E,Φ,A,Λ,O,H,O,H,Ψ,I,A,P,Φ,H,Λ,Γ,T,E,Φ,N,P,Γ,H,Ψ,O,A,P,I,N,Ψ,Γ,T,Λ,Φ,N,E,I,T,Ψ,O,H,P,I,A,E, O,T,Λ,Γ and **N.** 48 letters long was now the loop so this was also the part where I stopped.

This is the start per row 3→2→2→11→12→32

This is the loops per row 3→6→12→24→24→48.

The esoteric view.

I saw these patterns appearing in the first row but it disappears vastly after a couple of rows. Then I had the idea that the chaos must be expanding, by looking back at this it seems I was right and it seems logical that the chaos is expanding the deeper you work out the matrix. I would expect that if one should expand this until the 36th row that they will expand both exponentially according to some form of wave. Sadly my knowledge on number theory is limited and is this the best I can note.

So lets have a look at it from an different point of view, esotery.

When we look at the meaning of those numbers, 3 is divinity, 2 is duality, 6 is the macrocosm, 11 is the messenger/bridge, 12 is cosmic completion, 24 is the cosmic completion on the duality of dark and light. 32 are the paths of wisdom and 48 is the material temple.

The start is divinity → duality → duality → the bridge → cosmic completion → the paths of wisdom

The loop is divinity → macrocosm → cosmic completion → cosmic completion → the material temple

The behaviour of the start phase made me think of the Neo-Platonic and Hermetic stories of creation. This is a philosophical and mystical myth about how the cosmos and the human mind are made. Where modern math only sees chaos, this ancient story sees a logical journey.

The journey begins at the divine source, represented by the number 3. From there, the soul falls down into the physical world of duality, shown by the repeating number 2. This is like a spiritual pinball machine, where a person bounces back and forth between opposites—like light and dark, or good and bad—without truly moving forward. This is exactly why the beginning of the journey looks so chaotic.

To escape this trap, the traveller hits the number 11, which acts as a cosmic bridge. This bridge changes everything and allows the traveller to leave the chaos behind and move up to a higher spiritual level. After crossing it, the journey reaches the number 12, which stands for cosmic completion and natural order.

Finally, the path ends at number 32, which represents the paths of wisdom. At this exact point, the story is complete. The traveller has learned the rules of the universe and returned back to the divine source, which is why this journey reaches its perfect and natural end here.

Where the start phase describes the journey of the soul climbing back to the source, the loop phase describes the opposite path. It is the story of how the divine spirit steps down into the universe to create the physical world.

The creation begins at the divine source, represented by the number 3. From this pure point, the spirit expands outward into the number 6, which represents the macrocosm. This is the vast blueprint of nature, setting the cosmic directions of the universe.

As the creation solidifies, it reaches the numbers 12 and 24, which stand for cosmic completion and perfect balance. Here, the universe finds its natural rhythm, dividing the world into the harmony of light and dark, day and night. The double 24 shows that this cosmic order becomes deeply anchored and stabilized in the fabric of time.

Finally, the spirit reaches its destination at number 48, which represents the material temple. At this exact point, the invisible divine plan has fully densified into solid matter. The abstract ideas have become a physical home on earth. The creation is finished, and the spirit has successfully built its temple in the material world, which is why the pattern finds its natural resting place here.

Now with this part the length and frequency is seen in an esoteric lens but how about the form? We will sum everything up and use a module-11 to look what kind of archetype comes out from it.

- First row, start=3+6+9=(18-12)=**6.** And loop=(13-12=1)+5+(21-12=9)=**3**
- Second row, start=2+7=**9** and the loop =7+11+(15-12=3)+(19-12=7)+(23-12=11)+3=(((42-12=)30-12)=18-12)=**6**
- Third row, start=1+11=**12** and the Loop=17+15+13+11+9+7+5+3+1+23+21=**5**
- Fourth row, start=1+15+15+13+5+3+19+17+9+7+8=**4**
  Loop=3+7+17+21+7+11+21+1+11+15+1+5+15+19+5+9+19+23+9+13+23+3+13+17=**11**
- Fifth row, start=2+16+23+11+15+1+23+15+7+5+24+15=**1** and the
  Loop=5+9+1+11+21+13+17+15+13+17+9+19+5+21+1+23+21+1+17+3+13+5+9+7=**12**
- Sixt row, start=4+18+8+15+13+17+3+1+11+15+22+2+6+12+7+9+1+17
  +19+1+17+21+1+23+3+7+23+21+13+17+19+13=**10** and the
  Loop=5+21+1+11+15+7+15+7+23+9+1+17+21+7+11+3+19+5+21+13+17+3+7+23+15+1+17+9
  +13+23+3+19+11+21+13+5+9+19+23+15+7+17+9+1+5+15+19+11+3+13=**10**

So if we put it down nicely we see this.

Start. R1=6 R2=9 R3=12 R4=4 R5=1 R6=10
loop. R1=3 R2=6 R3=5 R4=11 R5=12 R6=10

But I was talking about archetypes, so what are the archetypes?

Start.	Loop.
6. ♍ Virgo - Earth	3. ♊ Gemini - Air
9. ♐ Sagittarius - Fire	6. ♍ Virgo - Earth
12. ♓ Pisces - Water	5. ♌ Leo - Fire
4. ♋ Cancer - Water	11. ♒ Aquarius - Air
1. ♈ Aries - Fire	12. ♓ Pisces - Water
10. ♑ Capricorn - Earth	10. ♑ Capricorn- Earth

The interesting thing is that the start in the first 6 rows act in perfect mirror symmetry all the while the loop acts with the opposites. 2 times air and earth and 1 time fire and air, but asymmetrically paired. There is one extra air and earth. I will put Diagram 1b from the book of signs below so the reader can understand the logica behind this.

DIAGRAM 1b.

Houses of the Zodiac					Rulers of the Houses			Periods referring to the Path of Return							Periods referring to the Path of the Fall	
Symbols	No's	Sanskrit Names	Latin Names	Elements of the Houses	The Twelve Primordial Titans of Greek tradition	Traditional Rulers: on the Day-throne	Traditional Rulers: on the Night-throne	Spheres of the Rulers	Traditional periods (based on the equinoxes and solstices of the Western and Sâyana tradition) (with very slight variations)	Traditional periods in the day-cycle (based on the birth-hour and not on astronomy)	Traditional periods in the life-cycle	Division with reference to the Four Elements, referring to the inner private life (beginning from the winter solstice)	Division with reference to the Four Elements of Matter, referring to the outer life in the world (beginning from the vernal equinox)	Four Âsramas (stages of life) of Hindu tradition (beginning from the Spiritual Pole)	Days of Genesis	Stages of the Fall (beginning from the Spiritual Pole)	Periods according to the Nirâyana Hindu System of Astrology (based on astronomy) (approximately from the middle of the months)
♑	I 10	Makara	Capricornus	Earth	Kronos (Saturnus)		Saturn	Earth	*Winter Solstice* 22 Dec.—20 Jan.	"*ghost-hour*" and "*Birth-hour*" 12—2 a.m.	*Incarnation or Conception* "pre-natal"				Evening of the First Day		January—February
♒	II 11	Kumbha	Aquarius (Fixed Sign)	Air	Ôkeanos (Oceanus)	Saturn		Earth	20 Jan.—18 Febr.	2—4 a.m. (3 a.m. is "zero-hour")	"pre-natal"	Earth	Embryo stage (Air)	midwinter (the Spiritual Pole)	Morning of the First Day		February—March
♓	III 12	Mina	Pisces	Water	Koios		Jupiter	Ether	19 Febr.—20 March	4—6 a.m.	"pre-natal"				Evening of the Second Day		March—April
♈	IV 1	Meśa	Aries	Fire	Krios	Mars		Fire	*Vernal Equinox* 21 March—21 April	*Hour of the Cock* 6—8 a.m.	*The Birth of Manifestation* from birth to 7 years			First (Earth)	Morning of the Second Day	Fourth (Earth)	April—May
♉	V 2	Vrishabha	Taurus (Fixed Sign)	Earth	Huperiôn		Venus	Water	21 April—21 May	8—10 a.m.	from 7—14 years	Water	Earth		Evening of the Third Day		May—June
♊	VI 3	Mithuna	Gemini	Air	Îapetos	Mercury		Air	21 May—21 June *Summer Solstice*	10 a.m.—12 *midday*	from 14—21 years *majority*				Morning of the Third Day		June—July
♋	VII 4	Karka	Cancer	Water	Theia		Moon	Moon	21 June—22 July	12—2 p.m.	from 21—28 years			Second (Water)	Evening of the Fourth Day	Third (Water)	July—August
♌	VIII 5	Simha	Leo (Fixed Sign)	Fire	Rheia	Sun		Moon	23 July—23 Aug.	2—4 p.m.	from 28—35 years	Fire	Water	midsummer	Morning of the Fourth Day		August—September
♍	IX 6	Kanyâ	Virgo	Earth	Themis		Mercury	Air	23 Aug.—23 Sept. *Autumnal Equinox*	4—6 p.m. *Evening Star*	from 35—42 years			Third (Fire)	Evening of the Fifth Day		September—October
♎	X 7	Tûla	Libra	Air	Mnêmosunê	Venus		Water	23 Sept.—23 Oct.	6—8 p.m.	from 42—49 years				Morning of the Fifth Day	Second (Fire)	October—November
♏	XI 8	Vrischikâ	Scorpio (Fixed Sign)	Water	Phoebê		Mars	Fire	23 Oct.—22 Nov.	8—10 p.m.	from 49—56 years	Air	Fire		Evening of the Sixth Day		November—December
♐	XII 9	Dhana	Sagittarius	Fire	Têthus	Jupiter		Ether	22 Nov.—21 Dec. *Winter Solstice*	10 p.m.—12 *Birth-hour*	from 56—63 years *The Birth of Light*			Fourth (Air)	Morning of the Sixth Day		December—January
♑	I 10	Makara	Capricornus	Earth	Kronos (Saturnus)		Saturn	Earth	22 Dec.—20 Jan.	12—2 a.m.	from 63—70 years				Evening of the Seventh Day (or First Day)	First (Air)	January—February
♒	II 11	Kumbha	Aquarius	Air	Ôkeanos (Oceanus)	Saturn		Earth	20 Jan.—18 Febr.	2—4 a.m. (3 a.m. is "zero-hour")	from 70—77 years or "after-death" (and "pre-natal")	Ether	Air	midwinter (the Spiritual Pole)	Morning of the Seventh Day (or First Day)		February—March
♓	III 12	Mina	Pisces	Water	Koios		Jupiter	Ether	19 Febr.—20 March	4—6 a.m.	from 77—84 years (or "after-death" and "pre-natal")				Evening of the Second Day (or Eighth Day)		March—April

I hope including this diagram also explains the earlier notions of the Hermetic and Gnostic roots. By looking at the table, you can see how the structural opposites and elements line up with the concept of finding balance within a closed system.

The thing that should be noted is that this is a rough interpretation of the knowledge I currently have, I need to look deeper into the sources to make a more accurate description of this but in basis this ought to be enough.

Please note that this is a rough interpretation of the knowledge I currently have. I need to look deeper into the sources to make a more accurate description of this, but in basis, this ought to be enough to see the framework.
The deeper meanings, connections, and ultimate conclusions are something I cannot definitively tell at this stage. These are simply the hard patterns I have observed, and this is written down to the best of my knowledge. I hope to make it more clear with time but I too are limited.

The thing is that this is now the first 6 rows. Even thought this is one cycle it does not mean the next one is exactly the same. For that the rest should be worked out and looked at the same way to check if it correlates. What is an interesting note though is that the startword and the starting chaos both are palindromes with 4,2,1,1,2,4 (Water, Earth, Fire, Fire, Earth and Water) and the word with 6,9,12,4,1,1 (Earth, Fire, Water, Water, Fire and Earth).

PS dear Paul Erdős, ‘’Mathematics may not be ready for such problems’’ prideful mathematicians that ignore things outside of their scope makes the Collatz conjecture impossible. When they grow up I’ll be sure humanity will be able to tame this dragon.

MadHatter out.

==========

LICENSING & COPYRIGHT NOTICE

==========

This documentation describes original mathematical hardware architecture and conceptual modulo-24 matrices.

Author/Licensor: Copyright © 2026 S. v Harten (MadHatter-Auditor). Some rights reserved.

This work is licensed under the strongly reciprocal open-source terms detailed below.

---

1. FOR THE MATRICES & HARDWARE ARCHITECTURE (CERN-OHL-S-2.0)

The embedded Greek Soyga Matrix configurations, design files, CAD blueprints, and any resulting physical or digital hardware implementations are licensed under the CERN Open Hardware Licence Version 2 – Strongly Reciprocal (CERN-OHL-S v2).

---

2. FOR SOFTWARE ARCHITECTURES & DERIVATIVE CODE (GNU AGPLv3)

Any software implementations, source code, scripts, AI model integrations, web applications, or digital tools derived from this work are licensed under the GNU Affero General Public License Version 3.

CRITICAL CONDITION: STRONGLY RECIPROCAL OBLIGATIONS

If you modify this documentation, implement these algorithm grids into software/network architectures, or construct hardware devices based on these files, you MUST make all modifications, derivative source files, code, and resulting hardware designs publicly available under the exact same respective license terms (CERN-OHL-S-2.0 for hardware/designs; AGPL-3.0-only for software and network services).

VERIFICATION & REFERENCES

The complete legal text of the licenses can be verified and downloaded at:

Hardware Steward: https://ohwr.org (SPDX: CERN-OHL-S-2.0)

Software Steward: https://gnu.org (AGPL-3.0-only)

Project Repository Reference: https://github.com/MadHatter-Auditor/vortex-anapole-confinement

==========

DISCLAIMER: THIS HARDWARE AND SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE LICENSOR BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES.

==========